Project: Worked examples
Title: Lifting lug and pin joint
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

A pinned lug between two cheek plates under a load along the lug: tension on the net section through the hole, bearing of the pin on the hole, shear tear-out beyond the hole, and the pin in double shear and in bending, each against an allowable stress; the largest utilisation governs.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Tension across the net section beside the hole | $\sigma_t = P/((W - d_h)\cdot t)$ |
| Bearing of the pin on the lug | $\sigma_{br} = P/(d_p\cdot t)$ |
| Tear-out shear beyond the hole, on two planes | $\tau_{to} = P/(2\cdot a\cdot t)$ |
| Pin shear, in double shear | $\tau_p = P/(2\cdot\pi\cdot d_p^2/4)$ |
| Pin bending, from the lever arm across the lug, the gap and the shackle | $M = (P/2)\cdot(t/4 + g + t_s/2)$; $\sigma_b = 32\cdot M/(\pi\cdot d_p^3)$ |
| Utilisation of each mode, its stress over the allowable from the material and design category in force (the sheet adds no design factor of its own) | $u = \sigma/\sigma_{allow}$ |

Basis: Pinned connections[1](Ch. 3); Shear, bearing and bending stresses[2](Ch. 3).

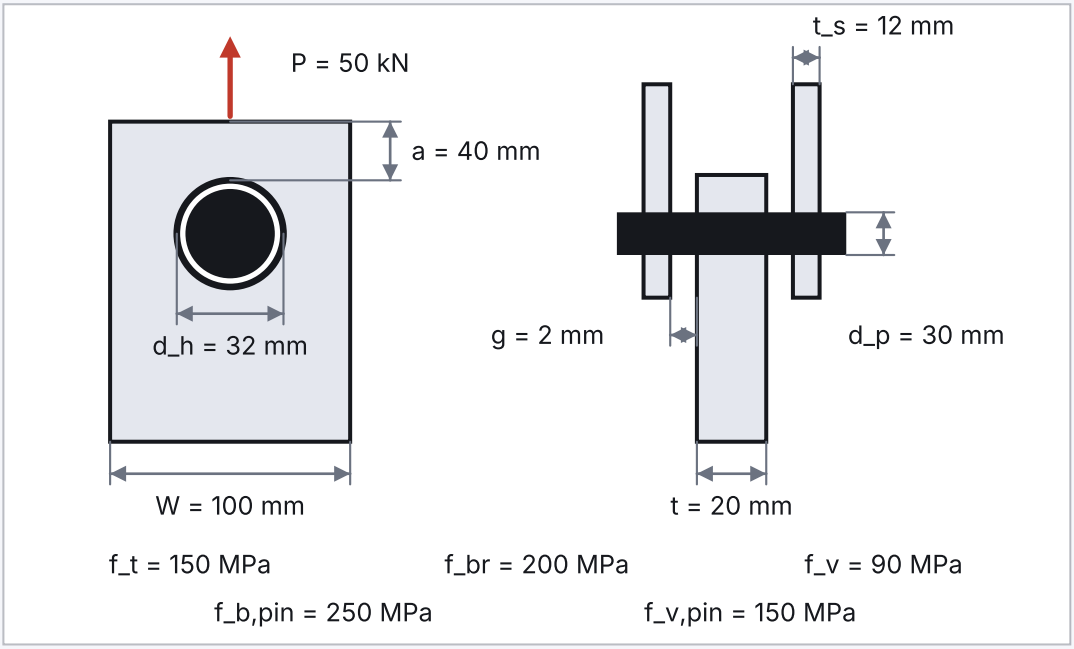



Figure 1 — A lug and its pin, front view and section through the pin: the lug between two cheek plates, the load through the pin. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| P | Load on the lug (along the lug) | 50 | kN | spec: LOAD-SCHEDULE-01 | 0.01 … 100000 kN |
| t | Lug thickness | 20 | mm | drawing: DRG-LL-101 rev A | 1 … 500 mm |
| W | Lug width at the hole | 100 | mm | drawing: DRG-LL-101 rev A | 5 … 2000 mm |
| d_h | Hole diameter | 32 | mm | drawing: DRG-LL-101 rev A | 2 … 1000 mm |
| a | Edge distance, hole edge to lug end (load direction) | 40 | mm | drawing: DRG-LL-101 rev A | 1 … 2000 mm |
| d_p | Pin diameter | 30 | mm | drawing: DRG-LL-101 rev A | 2 … 1000 mm |
| t_s | Cheek plate thickness (each) | 12 | mm | drawing: DRG-LL-101 rev A | 1 … 500 mm |
| g | Clearance, lug face to cheek plate (each side) | 2 | mm | drawing: DRG-LL-101 rev A | 0 … 100 mm |
| f_t | Allowable tensile stress, lug | 150 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 2000 MPa |
| f_br | Allowable bearing stress, lug | 200 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 3000 MPa |
| f_v | Allowable shear stress, lug | 90 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 2000 MPa |
| f_b_pin | Allowable bending stress, pin | 250 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 3000 MPa |
| f_v_pin | Allowable shear stress, pin | 150 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 2000 MPa |

## 3. Assumptions

A1. The load acts along the lug's axis through the hole centre. (No out-of-plane or side load.)
A2. The pin bears uniformly over the lug thickness and over each cheek plate; each cheek plate reacts half the load. (Pin-bending idealisation for a lug between two cheek plates.)
A3. A close-fitting pin (d_p ≥ 0.9·d_h). (Uniform-bearing idealisation.)

## 4. Working

```
[14] k_fit = d_p / d_h
= 30 mm / 32 mm
= 0.9375
```

Pin-to-hole ratio

[15] A_net = (W - d_h) * t
= (100 mm − 32 mm) × 20 mm
= 1360 mm²
Net section through the hole
[16] sigma_t = P / A_net [1](Ch. 3)
= 50 kN / 1360 mm²
= 36.765 MPa
Net-section tension
[17] A_br = d_p * t
= 30 mm × 20 mm
= 600 mm²
Bearing area
[18] sigma_br = P / A_br
= 50 kN / 600 mm²
= 83.333 MPa
Bearing of the pin on the hole
[19] A_to = 2 * a * t
= 2 × 40 mm × 20 mm
= 1600 mm²
Tear-out area (two planes beyond the hole)
[20] tau_to = P / A_to
= 50 kN / 1600 mm²
= 31.25 MPa
Tear-out shear
[21] A_p = 2 * pi * d_p^2 / 4
= 2 × π × (30 mm)^2 / 4
= 1413.7 mm²
Pin area in double shear
[22] tau_p = P / A_p
= 50 kN / 1413.7 mm²
= 35.368 MPa
Pin shear
[23] M_p = P / 2 * (t / 4 + g + t_s / 2)
= 50 kN / 2 × (20 mm / 4 + 2 mm + 12 mm / 2)
= 325 N·m
Pin bending moment (uniform bearing on lug and cheek plates)
[24] Z_p = pi * d_p^3 / 32
= π × (30 mm)^3 / 32
= 2650.7 mm³
Pin section modulus
[25] sigma_b = M_p / Z_p
= 325 N·m / 2650.7 mm³
= 122.61 MPa
Pin bending stress
[26] u_t = sigma_t / f_t
= 36.765 MPa / 150 MPa
= 0.2451
Net-section utilisation
[27] u_br = sigma_br / f_br
= 83.333 MPa / 200 MPa
= 0.41667
Bearing utilisation

[28] u_to = tau_to / f_v
= 31.25 MPa / 90 MPa
= 0.34722
Tear-out utilisation
[29] u_p = tau_p / f_v_pin
= 35.368 MPa / 150 MPa
= 0.23579
Pin shear utilisation
[30] u_b = sigma_b / f_b_pin
= 122.61 MPa / 250 MPa
= 0.49043
Pin bending utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| sigma_t | 36.765 MPa | Net-section tension |
| sigma_br | 83.333 MPa | Bearing stress |
| tau_to | 31.25 MPa | Tear-out shear |
| tau_p | 35.368 MPa | Pin shear |
| M_p | 325 N·m | Pin bending moment |
| sigma_b | 122.61 MPa | Pin bending stress |
| u_t | 0.2451 | Net-section utilisation |
| u_br | 0.41667 | Bearing utilisation |
| u_to | 0.34722 | Tear-out utilisation |
| u_p | 0.23579 | Pin shear utilisation |
| u_b | 0.49043 | Pin bending utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Pin bending | u_b = 0.49043 | u_b ≤ 1 | 0.49 | |
| PASS | Bearing | u_br = 0.41667 | u_br ≤ 1 | 0.417 | [1](Ch. 3) |
| PASS | Tear-out | u_to = 0.34722 | u_to ≤ 1 | 0.347 | [1](Ch. 3) |
| PASS | Net-section tension | u_t = 0.2451 | u_t ≤ 1 | 0.245 | [1](Ch. 3) |
| PASS | Pin double shear | u_p = 0.23579 | u_p ≤ 1 | 0.236 | |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.lifting-lug@1.0.1 Content hash 17ed215aa272cf35 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. American Society of Mechanical Engineers. Design of Below-the-Hook Lifting Devices. ASME BTH-1. ASME.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.